Sample Resume – Ph.D. Bioengineering

DENISE S. McGRAW

Elkridge, MD 21075 dsmcgraw@umd.edu
301-555-7890 in/dsmcgraw

EDUCATION

Ph.D., Bioengineering Anticipated May 2025
University of Maryland, College Park, MD
- Advanced to Candidacy, Nov. 2020
- GPA 4.0/4.0

B.S., Chemical Engineering May 2019
University of Maryland, College Park, MD
- Summa Cum Laude, with Engineering Honors
- GPA 4.0/4.0

RESEARCH EXPERIENCE

UMD Doctoral Dissertation Research Baltimore, MD
Researcher Jun. 2019 - Present

- Investigate poly (amido amine) dendrimers as oral drug carriers of anticancer therapeutics.
- Assess cytotoxicity, cellular uptake and transepithelial permeability of dendrimers and dendrimer-drug conjugates using *in vitro* Caco-2 cell model
- Independently determined impact of PEGylation of dendrimers on dendrimer transport, uptake and interactions with epithelial tight junctions.
- Co-wrote book chapter: R. Kolhatkar, **D. McGraw**, and H. Ghandehari, "Functionalized Dendrimers as Nanoscale Drug Carriers," in Multifunctional Pharmaceutical Nanocarriers, V. Torchilin (ed), Springer, 2012, pp. 201-232.
- Presented research poster at conference: **D. McGraw**, R. Kolhatkar and H. Ghandehari. "PEGylation of Anionic PAMAM Dendrimers: Implications for Oral Delivery." Poster presentation, 35th Annual Meeting of the Controlled Release Society, New York, NY, July 12-16, 2009.

UMD Undergraduate Research Project College Park, MD
Intern Aug. 2014 - Apr. 2016

- Completed a competitive, NSF-funded, Research Experience for Undergraduates (REU) summer internship program and then continued research project as an undergraduate research fellow.
- Determined the surface structure and chemistry of DNA-GaAs biochips using Grazing Incidence X-ray Scattering, X-ray Photoelectron Spectroscopy and Atomic Force Microscopy.
- Published manuscript in IEEE: M. Al-Sheikhly, **D. McGraw**, et al. "Radiation Induced Failure Mechanisms of GaAs Based Biochips," *IEEE Transactions on Device and Materials Reliability*. Vol. 4, No. 2., June 2005.

McGraw p.1/2

National Institute of Standards and Technology Gaithersburg, MD
Summer Undergraduate Research Fellowship (SURF) Jun.-Aug. 2015

- Developed an automated method to convert two-dimensional HIV protease inhibitor chemical structures to three-dimensional animations showcasing inhibitor interactions with protease active site using Pymol software.
- Created 300 visualizations of HIV Protease-Inhibitor interactions to supplement HIV research database (HIVSDB).
- Presented results to scientists and peers at SURF Symposium.

WORK EXPERIENCE

Graduate Teaching Assistant Sep. 2019 - May 2020
University of Maryland, College Park, MD

- Facilitated laboratory exercises for 40 students in Freshman Bioengineering Laboratory and delivered weekly lectures on engineering and biology topics.
- Graded 50 homework assignments weekly for Computational Methods in Bioengineering course and answered student questions concerning C and Matlab programming languages.

Undergraduate Teaching Fellow Jan. - May 2017
University of Maryland, College Park, MD

- Selected as a Women in Engineering Undergraduate Teaching Fellow.
- Developed lesson plans for and led a 1.5 hour recitation each week for 30 students in "Mass and Heat Transfer".

Clark School Ambassador Jan. - Aug. 2016
University of Maryland, College Park, MD

- Served as a student representative for the College of Engineering, giving presentations and tours to prospective students and parents, visiting local high schools and performing other recruitment duties.
- Developed curriculum and activities for "Discovering Engineering," a week-long engineering summer camp for middle school students.

LEADERSHIP AND EXTRACURRICULAR ACTIVITIES

- Tau Beta Pi, Recording Secretary (2014-2015) and Scholarship Chair (2016)
- Chemical Engineering Chair Search Committee, Undergraduate Representative (2013)
- Gemstone Coordinator Search Committee, Student Representative (2013)

AWARDS

- National Science Foundation Graduate Research Fellowship, 2019 – 2020.
- A. James Clark School of Engineering Dean's Award, May 2016.
- Barry M. Goldwater Scholarship, 2013 – 2014.

McGraw p. 2/2